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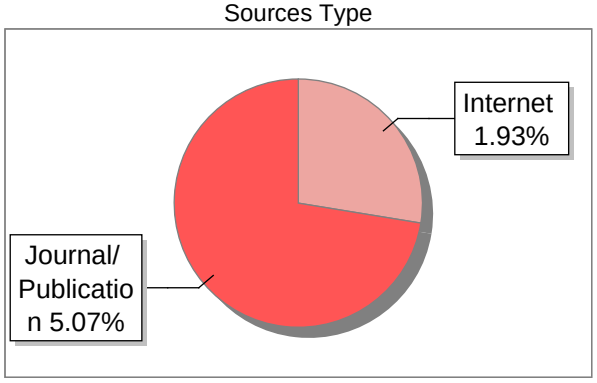

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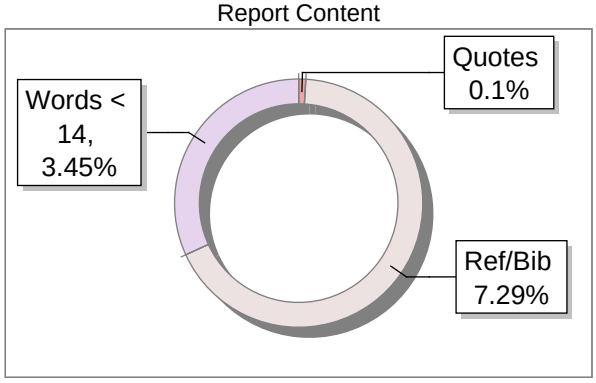

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ABSTRACT

VioTrack: A Deep Learning Approach for Violation Detection, Identity Extraction, and Digital Fine Management

Traffic offences remain one of the main sources of road hazards and urban 'bottlenecks'. Conventional enforcement approaches, such as visual loading by traffic personnel, are generally inefficient and subjective in nature, characterized by inconsistency between officers and subject to human errors and lagged processing time without consideration of the heaviness at dense urban area. As urbanization grows and the amount of vehicles on roads increases, smart automated systems with persistent monitoring for robust evidence based enforcement are urgently required.

VioTrack addresses this need by integrating deep learning–based violation detection, OCR-driven identity extraction, and automated digital fine management into a unified, scalable framework suitable for modern smart-city ecosystems. VioTrack employs an optimized computer vision model capable of detecting multiple types of traffic violations such as triple riding, riding without a helmet, and other rider-based violations through real-time frame analysis. The system uses a convolutional neural network (CNN)-driven detection pipeline that processes input video streams or surveillance images to accurately recognize violation events. Once a violation is detected, the framework's OCR module extracts vehicle registration details from license plates and relevant identity information from associated documents or text-bearing regions within the frame. This dual capability ensures that the enforcement process is not only automated but also verifiable and legally traceable. The extracted information is then passed to VioTrack's integrated Digital Fine Management System, which automatically assigns the appropriate penalty based on predefined rules and violation categories.

On the server side, evidence of violation photos with time stamps and extracted identification information and fine statistics are also safely stored. Concurrently, we are developing a Flask-based web application that allows enforcement officials to view detected violations on an interactive dashboard, validate extracted information manually as needed,

correct mistakes and issue digital receipts. It also has real-time support with features like payment confirmation samples, QR-based receipts, and (if enabled) email support so that the overall process will be faster, transparent and user-friendly.

A key strength of VioTrack lies in its design philosophy of practical deployment, focusing on efficiency and compatibility with widely accessible hardware.

Both detection and OCR models are designed to be CPU-friendly for real-time inference with low latency while still maintaining accuracy. Modular to easily add more complex features in the future, ie. multi-camera feeds, cloud-based computing, mobile notifications (notifications when a vehicle from a blacklisted database just entered an entry point of your city), integration with government databases. We perform extensive experimentation on collected down-sampled input image/video datasets including RGBD bottle/number plate samples to assess the fundamental performance of our system showing high accuracy in violation detection and accurate OCR extraction in different lighting, object distances as well as moderate occlusion conditions.

By automating the overall workflow – violation detection to identity extraction to fine issuance – VioTrack reduces manual effort, improves transparency, decreases human bias and helps reinforce road safety compliance. The flexible system architecture of the approach implies that it is scalable towards big-foot deployments in smart city infrastructures, it provides a feasible technological solution to contemporary traffic governance phenomenon. Focusing on accuracy, automation and user-oriented management, VioTrack serves as the basis of a full-service solution for traffic law enforcement digitalization.

2. INTRODUCTION

The managing of traffic in today's developing cities has become extremely complex than ever. As the number of vehicles are increasing each day, the cities has been crowded with these vehicles on the road and this has become the main issue on the cities for the traffic controlling officers to keep track of the records of the offences caused by many people through their vehicles on the roads each and every day.

Even though the government has told to follow the strict rules to follow on the places where many people still doesn't obey the rules to be followed which is provided by the government where by intentionally or unintentionally with their negligence of risk which they might face due to braking the road rules.

In the previous ways of monitoring or keeping track of offences was totally depended on the human which in other words can be understood as dependency on traffic controlling officers who used to control the offences manually by observing the roads, identifying the violations on time, and they used to record them manually with the available facility at that time.

So, knowing all these procedure are time consuming and less accuracy of data mainly during huge traffic where many offences might happen at the same made to overcome this issue by improving the facility provided to the government traffic officials by adopting new technologies such as Artificial Intelligence [AI] and vision of Computer has become a new evolution to track the violations or offences virtually. The cameras with the aid of AI models can look into the roads without break like humans and can recognize the type of violation on different situations just like humans could do. The Machine Learning [ML] helps to train the systems so that the systems will be able to react according based on the trained data and the situations can be like indicating the speed of the vehicles, the people riding the vehicles without helmet, not wearing seat belt and so on can be taken as the pattern to train and test the ML models to check their accuracy and can be implemented and makes sure that no offences or violations can be missed like they are missed through human eye. And when these techniques of AI and ML is combined it can be aided with the detection of the vehicle's number plated to recognize the registration number so that it aids to upload the offence over online so this make that the system is not just highlighting the violations it also identifying the vehicles too.

Here, the project "VioTrack" has been developed with the idea as a practical solution to support the authorities of the traffic offence controller's management. Which mainly focuses on the reduction of the traditional traffic monitoring method of controlling the traffic, keeping track of the offences and making them pay the fine of the offences which they have made by breaking the government rules. And this is possible to reduce the human dependency by making use of Artificial Intelligence and Machine Learning models can increase the efficiency of observing the roads through feeding camera and also can train by them by using

the various kinds of available data's like datasets and more with help of deep learning models to detect and process the real time data of offences.

This works like if the system finds any irrelevant activities, the system captures the entire frame, detects the registration number using the OCR technology, and all the related information's will be stored neatly on the government database. By storing all these details on the database aids the officials to review the data later, and to take the next action appropriately just by viewing the database instead of wasting their energy on the roads.

The agenda of this project is not to kill the human's jobs but to simplify their works from reducing the complexity. In short this project is just to reduce the work overload, inaccurate data but allowing the developed system to handle the works in an efficient way. In future, these systems might play a crucial role in reducing the rick but improving the safety on roads, decreasing the accidents and making sure that peoples follow the rules without breaking it.

As the cities are continuously developing, the issues also developing with that so to decrease these risk of issues we are developing the solutions with the help of new technologies to provide better way to manage the traffic and making the roads more safer than before to the society.

OBJECTIVES

The actual idea to implement this project is to create a practical system for the authorities of the traffic managers to notice the offences caused by breaking the violation rules automatically. Now, the monitoring of traffic is hugely dependent on the human's by observing manually. The manually refers to where the traffic managers on the road's has to watch the vehicles on the roads continuously, and take necessary action on real time, and also need to record it simultaneously which is really a burden to manage as a single person at that time. So, the purpose of the system is to reduce the pressure on the human with the help of current developing technology to handle the tasks which humans do in a systematic and organized way by maintaining accuracy of the data and reducing time.

The one of the aim of this project is the design a model which could analyse, detect and update the data of the violations with the fields like type of violation, fine amount, date of violation, time of violation along with the payment status. The different violations include triple riding, overspeeding, and overload and so on are often responsible for the cause of accidents, yet these are not notices always in all the real world scenarios. But by using the deep learning models which are already trained via supervised, unsupervised, semi-supervised and reinforcement techniques makes the system to observe the road without a small break and reacts to it very faster than human's.

The next crucial aim of this project is to extract the wordings from the number plate to identify the registration number of the vehicles to file the complaint of the offence they have made and to update the same on the database. These detection of violations and extraction on wordings and numbers from the number plates are possible only when the models are trained with help of OCR, algorithms, links and so on. This make the officers tasks easier to verify and update the vehicles offences without being depended on manual notes and collecting data manually.

Next this project along with focusing on extraction and detection it also focuses on storing those collected data in a systematic, organized, and structured format. Which aids in maintaining the report of violations and to retrieve the information for the officials inside the database whenever and wherever required. This will be the proper documentation which ensures its transparency and will be more reliable.

This Project also focuses on minimizing the manual efforts on officers. This doesn't need the officials to monitor the road as being physically present in all the roads. This system is used as a support layer, to watch the areas to control the traffic without breaks. This aids the managing authorities to manage the time and to focus on solving the critical issues of traffic.

Finally, the last objective is to develop a system which simplifies the use and can be developed for the future enhancements. This is the current version which focuses on the basic traffic violations, but this development can be improved by including more advanced detection techniques in future. This system also provides the flexibility of limitless to a particular function.

4. SCOPE

The scope of this project is all about the implementation of a simple and practical system which can detect the common violations of traffic automatically without any kind of manual input but it takes inputs using camera and machine learning. The main focus is on violation identification such as riding without helmet, signal jump, and other similar violations which compromise the safety on roads. If this system doesn't try to handle all possible scenarios in traffic, it at least tries to work for the selected violations and provides and accurate and proper results.

Even though this project provides the limited working functionality with the images or videos captured input in the basic normal lighting conditions. It is not able to deal with heavy complex environments like weather during extreme times, lighting which is very low, or scenes where huge crow causes poor visibility. The aim is to build the working model which performs well on the standardized conditions which are commonly found on the junctions and roads etc.

The next scope of this project is involves the extraction of numbers and wordings from the number plates. Whenever a violation is detected, the system is assigned to extract the vehicles number plate to extract its registration number using the OCR technology. This OCR technology helps to link the violations with the respective vehicle. However, the assigned performance to the OCR depends on the clarity of the inputs it can be either image or video etc. If the number plates are damaged, covered, or too unclear which means blurred may not generate the perfect output. Even though if the system faces the above mentioned scenarios it tries to handle the normal cases efficiently.

This project also involves the creation of database to store and retrieve the information of detected violations and its details. This aids the officials to access the data anytime and anywhere required to them. Here the database can be considered as the backbone of the developed system as it ensures that all the report are recorded properly and it maintains its consistency without being altered to make sure nothing will be lost. This can be enhanced in further future when additional fields need to be added and when different violations required according to future developments.

The project “VioTrack” doesn’t focus only on the automatic fine generation or to communicate directly with the drivers. Instead, it focuses on detecting violations, and keeping records, which are the really basic steps when required before any further fields added through animations. In the future, the aid is to improve the support for the traffic management by adding the advanced features based on the future requirements such as sending the alert messages, generating reports, or combining the system enforcements.

Overall, the scope of this project is designed to keep this project as simple as possible, achievable, and makes it useful for the real life situations. It also provides a strong foundation which helps to develop further while this project is already offering the support as much as possible to traffic authorities for identifying and recording accurate violations and its details.

5. LITERATURE SURVEY

Before diving into the implementation of this project, it was very crucial to understand how various researchers and the developers have approached the issues of road traffic violation detection and automated monitoring. As per the many studies it shows that traffic enforcement even in this huge traffic cities are depended on manual supervision, which makes the monitoring complex or sometimes unseen, especially in huge traffic areas. Because of this, many researchers have gone through many ways to use the computer vision, artificial intelligence and machine learning models to make these processes more reliable and less dependency on humans.

The early researches are primarily focused on detecting numbers and wordings from the number plates using the traditional technique for image processing. Those systems used the edge detection, methods of contour, and they used threshold to locate the number plates for detection of numbers and wordings on it. Even though these methods worked to some extent, they too struggled due to the lighting issues like high or low, the plate was not in the appropriate angle, or due to the low quality of the camera which used to capture the inputs of violations. Because of these issues, the results of the traditional methods was not accurate.

When deep learning has started to become more popular as become common in the current developing technologies, many studies are also started using the Convolutional Neural Networks (CNNs) which is used for object detection. Models such as YOLO (You Only Look Once) which generally used, SSD (Single Shot Detector), and Faster R-CNN also became every popular because they are used to identify the objects quickly and accurately which is suitable to capture the moving vehicles. Researchers also found that these developed models are performed well for detection of various violations like detection of riding without helmet, detection of triple riding, and also vehicle classifications. According to the research most papers has reported an improvement in the accuracy when they have replced the deep learning models with the traditional approaches of image processing.

Recently many research work also discussed about the OCR (Optical Character Recognition) for reading the wording and numbers from the number plates which are the registered number of the vehicles. The OCR tools like Tesseract has been considered as most widely used because they free of cost, easy it use, and provides decent outputs when the number plate is clear. However, many studies shows that OCR is still struggling with the images which are unclear, number plates with different fonts, the number plates which are bent and filled with dirt and frames with low resolutions. Because of this, the researchers have sometimes combined the OCR with pre-processing techniques to improve the readability of the number plates.

Another area which appears more frequently in the literature is the management of the database for the enforcement of traffic. As many systems stores the violations with the images, locations, timestamps, and vehicle details so that it becomes easy for the officers to view it later. Even some studies has also explored automating fines, but they usually require strong accuracy and government approval, so here this project is for the academic purpose so it mainly focuses on the detection rather focusing on the fine generation and doing payment.

Overall, based on the researches most of the research agrees to that deep learning is one of the best way to detect the violations of traffic automatically. As compared to the previous developments this project handles the various lighting conditions better, adjusts to different angles when the number plates on inappropriate angles, and provides faster results than ever before compared to existing systems. However, the researchers also points out the challenges such as the requirements of the training data, accuracy of the model in the real world

environments, and the requirement of the hardware which is needed to handle the processing video smoothly without any issue or interrupt.

This “VioTrack” project is built with the help of those finding successfully. The implementation ideas are from the previous studies which helped shape the final design by showing the best techniques which are reliable based on the requirement of the project and which limitations is still exist. So, instead of trying to solve all the problems everything at once, this project focuses on divide and conquer method which can be referred as focusing on small set of violations rather than all and tries to create a simple but effective system development based on the lessons learned from the earlier researches.

6. EXISTING SYSTEM

Now, in most of the cities the monitoring of the violations of traffic is still handled mainly through traditional way which is the manual observation. Where the traffic police stands in the junctions, road or in the centre, watches closely, and tries to identify the people who are not following the basic rules. Since this method has been followed from many years, it has various limitations which are making the enforcement complex and inconsistent. The human officers can also focus on one area at a time, and during those busy hours it actually becomes almost impossible to keep track on each and every vehicle and each and every violations that has being taken place around them at the same time.

The next problem with the manual traffic monitoring is about dependency on the officer’s attention during controlling of traffic and the ability the react to those violations quickly. If the traffic is huge too many vehicles passes at the same time or if the visibility is poor, some of the violations might not get noticed. Even when a violation is identified, the recording details such as the vehicle number plate, time, and type of violations takes more time to update it manually. As human mistakes can happen during these processes, mainly during the stressful or also in the fast moving situations.

In few areas, the CCTV cameras have been installed, but they are most of the times used for the surveillance purposes only where using the footages they update the violations inside

their databases and not for the automatics detection where it detects the violations and updates its data automatically. In other words, the government traffic officers need to review the CCTV footages manually from the recordings, which is actually time consuming and not very possible thing to make all the time. The traditional cameras does not provide any features like alerts in real time or extraction of number plates to extract the numbers and wordings automatically which decreases the workloads and the other working remains same like old traditional method but it will be automated.

Few systems tries to make use of the regular methods for image processing for detections, but these methods are not well suitable for the real realm conditions. Why because those traditional methods will get affected by the changes in the images like lightings, shadows, vehicle angles, number plate angles, and also low quality of the images for processing it. Because of these kinds of issues or challenges, these techniques of traditional methods cannot be reliable to detect violations or read number plates accurately.

Hence, as overall it is understood that the existing systems are not much efficient for managing and detecting today's traffic. So, with these increasing number of vehicles on the cities, the traditional manual monitoring has become less effective and few times it's leading to less accuracy in collecting the violations or delays in taking actions for the violations. And these situations has made us to implement the new automation of identifications, recording the process, detection, and focusing more on resolving the issues rather than just monitoring continuously ideas to support the offence management officials.

7. PROPOSED SYSTEM

This proposed system is designed to decrease the dependency on human for traditional way of manual monitoring of traffic to identify, detect, and take action based on the violations as much as faster and as more reliable. So, what this system does is instead of expecting the government officials to monitor the roads without any breaks, the system uses the cameras to get the input to monitor it automatically along with the trained deep learning model with is used to detect the common violations automatically. The main agenda is to build a system

which works in the background, monitors the roads in real time without breaks, and provides the alerts to the authorities to take actions whenever a violations takes place.

In this developed system, the camera is used for video streaming which was to process the frame by frame. This machine learning model is trained in such a way that it must be able to recognize the specific situations. For example, riding without helmet or triple riding and more where we can notice that this system is capable to analyse the each frames. Whenever the model detects the violation, it marks that area where the violation had taken place or occurred and stores that frame of violation for the further processing of violation actions. This aids in reducing the need for the human officers to review the long period of CCTV footages manually which is time consuming.

When a violations is detected by the system, its next step is to extract the number plate where it extracts the government registration number by analysing the numbers and wordings of the number plates. To perform these operations, the system will be using the OCR to read the text on the vehicles number plates. The next condition is if the number plate is not in the proper way like filled with dirt or inappropriate angle the system will be unsuccessful to detect it. But if the number plate is clear and the image is captured properly, the system will be able to extract the vehicle number plates accurately. And all the extracted details like which includes the type of violations, time of violations, image of violations, and these details will be stored inside the database. This methods of storing the details inside the database makes it easier to access those data for the officers to loo up the records stored whenever they want.

The one of the main aim of this proposed system is to make these processes simple and easily understandable. The user interface is designed in such a way that so the officers can easily log in, check the violations, and allows to verify the information without the need of any technical knowledge which means a person with basic knowledge can also access and use this system. This system is built in such a way that it is flexible which means it can be easily expanded if required in the future. Like more number of violations can be added, can improve the detection model, and can also include the feature like automated fine generation in the future.

The next advantage of this system is that it works continuously without getting distracted or tired. And also it does not depend on any things like flow of traffic and conditions of weather as long as the input method which is the camera is functional, the system continue to work

without any breaks and issues. This provides the stronger support to the traffic managing authorities and aids them to maintain the order more efficiently and effectively.

At last, the proposed system's agenda is to provide the support through automating the traffic monitoring, reduces the efforts of human dependency, and increases the detection of violation. With the combination of machine learning, OCR, database management, this system is providing the complete solution which can aid in improving the safety of roads and making the enforcements more organized.

8. METHODOLOGY

The methodology of the system explains that how the system works in each step as a step by step procedure, which is starting from gathering the video input for detecting the violations and till storing the final results. The each part of this process is kept as simple as possible so that it makes sure that the system runs or functions smoothly without any issues and the officers can easily understand how the detection is happening. This complete workflow is completely divided in several stages, and each stage is contributing to the final outcomes of identifying the violations of traffic automatically.

The primary stage is data collection which is about gathering the required data, where the inputs like video and image is taken from the camera. The camera can be placed at a junction or at any point where monitoring is needed so that the input can be clear enough to update the violation details inside the database. The quality of the input is the most major thing to focus on because the model is mainly depended on clear frames to identify and detect the violations accurately. The system will able to read one frame video at a time and also prepares the other frames for analysis by altering its size and adjusting its size to satisfy the model's conditions.

The second stage is, detection of violation using the deep learning model. Which is a pre-trained model specially designed to recognize the particular violations like not wearing helmet or triple riding, signal jump, over speeding is used here for this system. This developed model is implemented in such a way that is must scan each and every frame and also checks if any violation is present or not. When this model identifies something, it marks a bounding box around the detected frame area and marks it for further future processing.

This part is the very crucial for the system because the accuracy of the detection is totally depended on how well this model has been trained.

Once the system is detected a violation, the system moves to the next phase or the next stage, which is the extraction of number plate. In this phase, the frame area where the vehicle's registered number plate is located in cropped and further processed through OCR technology. The OCR technology is used to attempt the extraction in other words it tries to read the characters, numbers, or wordings from the number plates and later converts them into text. The success of this step or phase is completely depended on how clear the number plate is, the lighting conditions, the angle of the number plate, and the angle at which the vehicle is captured. If the text is in the readable format, the system returns the required details based on the taken input from the number plates.

The next phase or step is about storing the results of detection. All the information such as the type of violations, vehicle registered number, time of violation, the frame captured and all other details will be recorded in a database. This aids to maintain and record the details so that those data can be easily retrieved whenever and wherever needed. The officers can log in at any time and view all violations in a table format that too with proof of images in a clear and organized way, which makes it easier for the management of offences to verify and take further actions.

The next part of the methodology includes designing of simple interface for user's usage. The system is designed with the login page and a section where all the violations can be viewed all in one place. This interface is built to be user-friendly so that the user with or without basic or proficient technical background and knowledge can understand and access the recorded details easily without any difficulty.

The final stage or phase of the system is testing and refining, this is to ensure that the system works under various kinds of situations. All the required adjustments are already made to improve the accuracy of detection, to handle the number plates which are unclear, and also to manage different situations of lightings. The aim is of this stage or phase is to ensure that the system's performance is reliable in real life situations and not for in ideal conditions.

In short, this methodology brings the combination of computer vision, OCR, and database management to form a complete solution compare to old traditional methods. The each and

every stage is connected, and the flow also ensures that violations are identified, detected, and saved in a proper format, also reducing the manual work and supports the traffic authorities efficiently and effectively.

9. CONCLUSION

This “VioTrack ” project is developed with the intension to create a simple, effective and efficient system which can aid traffic officers or authorities to identify the people who is breaking the rules of violations by monitoring automatically. As we are seeing this rapid growth in the in the increasing number of vehicles on each cities, it has become extremely harder or complex for the officers to handle and manage the traffic in every junction and to take action to each traffic violating person’s. With this combination of machine learning, OCR, and along with the database management, this system is providing a way to decrease the workload of the officers and to make the monitoring process easy and organized.

Throughout the designing and development of this “VioTrack” project, it has become a clear that the deep learning models plays a crucial role in providing the safety on roads. Here, this model is used to make the selected violations detectable such as triple riding, riding without helmet, over speeding with the reasonable accuracy. While this system is not flawless and may sometimes might miss few violations or can become unsuccessful to extract the data from the number plates when it’s not clear, it still provides a reliable starting point for automated monitoring facility. Along with the more number of training data and with the better cameras with quality, these outcomes can become even stronger.

The one of the crucial outcomes to focus on this project is the way it brings together various components into the one working system. Here, the detection model, which is the OCR model, along with the database all works in a linear way to identify a violation through camera inputs and to store the details properly inside the database. These techniques are making the tasks easier for the officials for checking the records instead of scanning the long-time video footages manually. Here, the user interface is also helping in presenting the information in a formatted way such that making it easily understandable.

The another key point learned during the development of this project is that the automations used in this project does not going to replace human officers but killing their works it's just supporting them by simplifying their task automatically instead of using the traditional methods. The system will be working in such a way that it works without any breaks and can monitor the areas where the officers cannot focus and take action closely. By decreasing the manual efforts for checking, the officers can focus more on solving the issues of roads and managing the traffic better.

Overall, this project represents that automated traffic monitoring can be very useful feature to the existing methods of enforcement. This simple is simple, practical, and can be explained in various ways. Along with other further improvements, such as handling other more additional types of violations and to improve the accuracy of the data extraction from the government registered number plates, this approach can contribute to a roads safety and provides more

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15. Google Scholar was used to explore existing studies on AI-based traffic monitoring and rule-enforcement systems.
16. Gonzalez and Woods' *Digital Image Processing* (2008, Pearson) served as a reference for basic and advanced image-processing concepts.
17. Adrian Rosebrock's *Deep Learning for Computer Vision* (PyImageSearch, 2019) was helpful for understanding practical computer-vision workflows.